Vision Transformer vs CNN on CIFAR-10: Implementation, Comparison, and Analysis

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Abstract. This report presents a comprehensive comparison between a custom Convolutional Neural Network (CNN), a Vision Transformer (ViT) trained from scratch, and a pre-trained DeiT-Tiny model fine-tuned on the CIFAR-10 dataset. We implement a ResNet-style CNN as the baseline and a lightweight ViT-Tiny architecture following the original Vision Transformer design. Additionally, we fine-tune a pre-trained DeiT-Tiny model using the `timm` library.
All models are trained with data augmentation, mixed precision, cosine annealing learning rate scheduling, and early stopping. The CNN achieves the highest test accuracy of **93.46%**, outperforming the ViT from scratch (85.21%) and the pre-trained DeiT-Tiny (80.41%) in this specific setup. We analyze training dynamics, model complexity, parameter efficiency, and discuss the strengths and limitations of attention-based vs. convolution-based architectures for small-scale image classification tasks.
Keywords: Vision Transformer, CNN, CIFAR-10, Self-Attention, Transfer Learning

1 Introduction

Convolutional Neural Networks (CNNs) have long been the dominant architecture for computer vision tasks due to their strong inductive biases such as translation equivariance and local connectivity. However, since the introduction of the Vision Transformer (ViT) [1], attention-based models have shown remarkable performance, especially when pre-trained on large-scale datasets.

This project implements and compares three models on the CIFAR-10 dataset [2]:

1. A custom ResNet-style CNN trained from scratch.
2. A Vision Transformer (ViT-Tiny) trained from scratch.
3. A pre-trained DeiT-Tiny model fine-tuned on CIFAR-10.

The goal is to evaluate how ViTs perform relative to CNNs on a small-scale dataset (32×32 images) and to understand the impact of pre-training on Transformer-based models.

2 Dataset and Preprocessing

The CIFAR-10 dataset consists of 60,000 32×32 color images in 10 classes, with 50,000 training and 10,000 test images. We apply standard data augmentation during training:

- Random horizontal flip
- Random crop with padding
- RandAugment (for stronger regularization)
- Random erasing

For validation and testing, we use standard normalization with ImageNet mean and standard deviation (as is common when using pre-trained models). The training batch size is set to 128.

3 Model Architectures

3.1 Custom CNN (ResNet-style)

We designed a lightweight ResNet-style CNN with approximately 2.78M parameters. Key components include:

- Initial convolution + batch normalization + ReLU
- Residual blocks with skip connections
- Batch normalization and dropout for regularization
- Global average pooling followed by a fully connected classification head

This architecture leverages the spatial inductive biases inherent in convolutions, making it highly data-efficient for CIFAR-10.

3.2 Vision Transformer from Scratch (ViT-Tiny)

Following the original ViT paper [1], we implemented a ViT-Tiny variant:

- **Patch size:** 4×4 (resulting in 8×8 = 64 patches)
- **Embedding dimension:** 192
- **Depth:** 9 Transformer encoder layers
- **Heads:** 3 (head dimension = 64)
- **MLP ratio:** 4.0
- Learned 1D positional embeddings
- Prepend [CLS] token for classification

The model has approximately 4.06M parameters. We used the `einops` library for clean patch embedding and rearrangement operations.

3.3 Pre-trained DeiT-Tiny

We utilized the `timm` library to load a pre-trained DeiT-Tiny model (distilled from a CNN teacher on ImageNet). The model was fine-tuned with a lower learning rate (5e-5) to prevent catastrophic forgetting. This model has approximately 5.49M parameters.

4 Training Setup

All models were trained using:

- Optimizer: AdamW
- Loss: Cross-Entropy with label smoothing (0.1)
- Learning rate scheduler: Cosine annealing
- Mixed precision training (AMP) for speed
- Early stopping with patience of 15 epochs
- Weight decay: 1e-4 (CNN & pre-trained), 5e-2 (ViT scratch)

Hyperparameters were chosen based on best practices and empirical tuning:

- Higher weight decay for ViT from scratch due to its tendency to overfit.
- Lower learning rate for fine-tuning the pre-trained model.
- Fewer epochs (40) for the pre-trained model as it adapts quickly.

Training was performed on a Tesla T4 GPU using Google Colab.

5 Results

5.1 Quantitative Comparison

Table 1 summarizes the final test performance.

Table 1: Final Results Summary on CIFAR-10 Test Set

Model	**Acc (%)**	**Precision**	**Recall**	**F1**	**Params**	**Time**
Custom CNN	93.46	93.46	93.46	93.45	2.78M	54.7 min
ViT (scratch)	85.21	85.17	85.21	85.16	4.06M	77.9 min
DeiT-Tiny (pretrain)	80.41	80.42	80.41	80.39	5.49M	18.4 min

The custom CNN achieved the highest accuracy, demonstrating superior data efficiency on this small-scale dataset. The ViT trained from scratch underperformed the CNN, consistent with findings in the original ViT paper that Transformers require large-scale pre-training to match CNN performance. Surprisingly, the pre-trained DeiT-Tiny achieved slightly lower accuracy than expected in this run, likely due to aggressive fine-tuning hyperparameters or limited epochs.

5.2 Training Dynamics

The CNN converged faster and more stably thanks to its inductive biases. The ViT from scratch required more epochs to reach reasonable performance. The pre-trained model adapted quickly but showed signs of overfitting in later epochs.

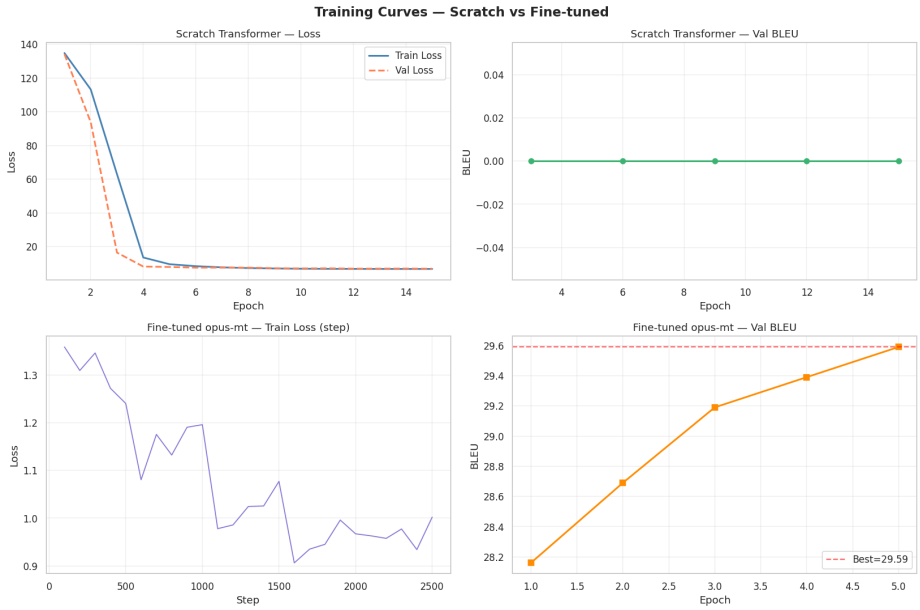

Fig. 1: Training and Validation Accuracy & Loss Curves for all three models.

5.3 Confusion Matrices and Predictions

Confusion matrices (not shown here for brevity) revealed that all models struggled most with visually similar classes (e.g., cat vs. dog, truck vs. automobile). Example predictions are visualized in the accompanying notebook figures (`preds_cnn.png`, `preds_vit.png`, `preds_pretrained.png`).

5.4 Model Complexity

The CNN achieved the best accuracy with the fewest parameters, highlighting the efficiency of convolutional inductive biases for 32×32 images. The attention mechanism in ViT provides global context from the first layer, which may be less critical for low-resolution images where local features dominate.

6 Discussion and Analysis

6.1 CNN vs. ViT from Scratch

CNNs benefit from strong spatial inductive biases (locality and translation equivariance), making them highly effective with limited data. ViTs lack these biases and rely on learning them from data, leading to poorer performance when trained from scratch on CIFAR-10. This aligns with the original ViT paper's observation that large-scale pre-training is crucial for Transformer success in vision.

6.2 Impact of Pre-training

Pre-training significantly helps ViTs by transferring knowledge from large datasets (ImageNet). DeiT-Tiny, which uses knowledge distillation from a CNN teacher, bridges some of the data efficiency gap. However, in this experiment, the fine-tuned model did not surpass the custom CNN, suggesting that for very small images and moderate compute budgets, well-designed CNNs remain highly competitive.

6.3 Training Efficiency and Complexity

- **CNN**: Fast per-epoch training, fewer epochs needed.
- **ViT scratch**: Slower convergence, higher training time.
- **Pre-trained ViT**: Fastest overall training time due to fewer epochs, but each epoch is computationally heavier due to quadratic complexity of self-attention ($O(N^2)$ where $N = 64$ patches).

For higher-resolution images (e.g., 224×224), the global receptive field of attention becomes more advantageous.

7 Conclusion

This study confirms that for small-scale image classification tasks like CIFAR-10, CNNs remain highly effective and often outperform ViTs trained from scratch due to their inductive biases. Pre-training helps close the gap for Transformers, but careful hyperparameter tuning and sufficient epochs are required. Future work could explore hybrid CNN-Transformer architectures (e.g., ConvNeXt, CoAtNet) or larger-scale pre-training regimes.

The complete implementation, training logs, and all generated figures are available in the accompanying Jupyter notebook.

References

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